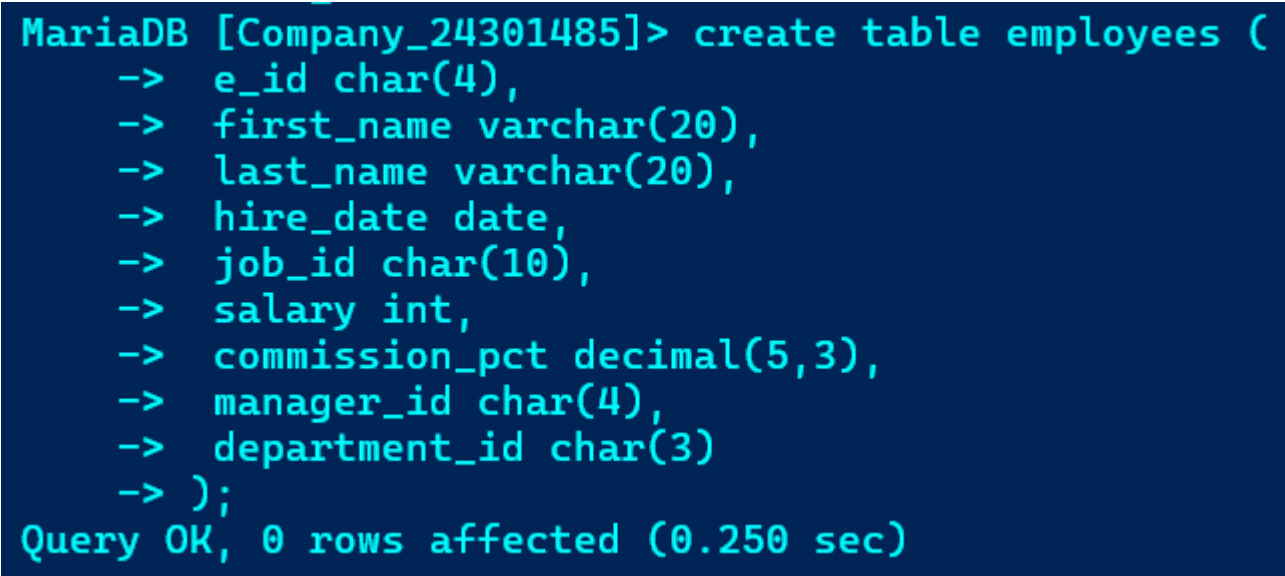


CSE370 : Database Systems

Assignment 02 | Summer 2026

ID : <24301485> | Name : Taufiq Mustafizur Rahman

No 1 Query (as Plain Text)	<pre>create table employees (e_id char(4), first_name varchar(20), last_name varchar(20), hire_date date, job_id char(10), salary int, commission_pct decimal(5,3), manager_id char(4), department_id char(3));</pre>
No 1 SS (of Query & Output in Shell)	 <pre>MariaDB [Company_24301485]> create table employees (-> e_id char(4), -> first_name varchar(20), -> last_name varchar(20), -> hire_date date, -> job_id char(10), -> salary int, -> commission_pct decimal(5,3), -> manager_id char(4), -> department_id char(3) ->); Query OK, 0 rows affected (0.250 sec)</pre>

No 2 Query (as Plain Text)	INSERT into employees values ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'), ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');

No 2 SS
(of Query & Output
in Shell)

```
MariaDB [Company_24301485]> INSERT into employees values
-> ('E001', 'John', 'Doe', '2020-01-15', 'IT_PROG', 65000, NULL, 'E010', 'D01'),
-> ('E002', 'Jane', 'Smith', '2019-03-22', 'SA_REP', 55000, 0.150, 'E011', 'D02'),
-> ('E003', 'Michael', 'Johnson', '2021-06-10', 'IT_PROG', 62000, NULL, 'E010', 'D01'),
-> ('E004', 'Emily', 'Davis', '2018-11-05', 'SA_REP', 58000, 0.200, 'E011', 'D02'),
-> ('E005', 'David', 'Wilson', '2022-02-14', 'IT_PROG', 60000, NULL, 'E010', 'D01'),
-> ('E006', 'Sarah', 'Brown', '2020-07-30', 'HR_REP', 50000, 0.150, 'E012', 'D03'),
-> ('E007', 'James', 'Taylor', '2019-09-18', 'SA_REP', 60000, 0.180, 'E011', 'D02'),
-> ('E008', 'Lisa', 'Anderson', '2021-04-25', 'FI_ACC', 52000, NULL, 'E013', 'D04'),
-> ('E009', 'Robert', 'Martinez', '2020-12-01', 'IT_PROG', 67000, NULL, 'E010', 'D01'),
-> ('E010', 'Patricia', 'Garcia', '2017-05-10', 'IT_MAN', 85000, NULL, NULL, 'D01'),
-> ('E011', 'Thomas', 'Lee', '2016-08-20', 'SA_MAN', 80000, 0.100, NULL, 'D02'),
-> ('E012', 'Maria', 'White', '2018-02-15', 'HR_MAN', 75000, NULL, NULL, 'D03'),
-> ('E013', 'Daniel', 'Harris', '2019-10-30', 'FI_MAN', 78000, NULL, NULL, 'D04'),
-> ('E014', 'Jessica', 'Clark', '2021-01-20', 'HR_REP', 48000, 0.100, 'E012', 'D03'),
-> ('E015', 'Kevin', 'Lewis', '2022-03-12', 'FI_ACC', 51000, NULL, 'E013', 'D04');
```

Query OK, 15 rows affected (0.082 sec)

Records: 15 Duplicates: 0 Warnings: 0

```
MariaDB [Company_24301485]> Select * from employees;
```

e_id	first_name	last_name	hire_date	job_id	salary	commission_pct	manager_id	department_id
E001	John	Doe	2020-01-15	IT_PROG	65000	NULL	E010	D01
E002	Jane	Smith	2019-03-22	SA_REP	55000	0.150	E011	D02
E003	Michael	Johnson	2021-06-10	IT_PROG	62000	NULL	E010	D01
E004	Emily	Davis	2018-11-05	SA_REP	58000	0.200	E011	D02
E005	David	Wilson	2022-02-14	IT_PROG	60000	NULL	E010	D01
E006	Sarah	Brown	2020-07-30	HR_REP	50000	0.150	E012	D03
E007	James	Taylor	2019-09-18	SA_REP	60000	0.180	E011	D02
E008	Lisa	Anderson	2021-04-25	FI_ACC	52000	NULL	E013	D04
E009	Robert	Martinez	2020-12-01	IT_PROG	67000	NULL	E010	D01
E010	Patricia	Garcia	2017-05-10	IT_MAN	85000	NULL	NULL	D01
E011	Thomas	Lee	2016-08-20	SA_MAN	80000	0.100	NULL	D02
E012	Maria	White	2018-02-15	HR_MAN	75000	NULL	NULL	D03
E013	Daniel	Harris	2019-10-30	FI_MAN	78000	NULL	NULL	D04
E014	Jessica	Clark	2021-01-20	HR_REP	48000	0.100	E012	D03
E015	Kevin	Lewis	2022-03-12	FI_ACC	51000	NULL	E013	D04

15 rows in set (0.001 sec)

No 3 Query (as Plain Text)	Select first_name, last_name, hire_date, department_id from employees WHERE hire_date = (Select MAX(hire_date) from employees);
No 3 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select first_name, last_name, hire_date, department_id from employees WHERE hire_date = (Select MAX(hire_date) from employees) +-----+-----+-----+-----+ first_name last_name hire_date department_id +-----+-----+-----+-----+ Kevin Lewis 2022-03-12 D04 +-----+-----+-----+-----+ 1 row in set (0.003 sec) </pre>
No 4 Query (as Plain Text)	Select first_name, last_name, e_id, salary, department_id FROM employees e1 WHERE salary = (Select MAX(salary) FROM employees e2 WHERE e1.department_id=e2.department_id);
No 4 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select first_name, last_name, e_id, salary, department_id FROM employees e1 WHERE salary = (Select MAX(salary) FROM employees e2 WHERE e1.department_id=e2.department_id); +-----+-----+-----+-----+-----+ first_name last_name e_id salary department_id +-----+-----+-----+-----+-----+ Patricia Garcia E010 85000 D01 Thomas Lee E011 80000 D02 Maria White E012 75000 D03 Daniel Harris E013 78000 D04 +-----+-----+-----+-----+-----+ 4 rows in set (0.001 sec) </pre>
No 5 Query (as Plain Text)	Select first_name, last_name, e_id, commission_pct, department_id from employees WHERE department_id='D02' and commission_pct > ALL (Select commission_pct FROM employees WHERE department_id = 'D01' AND commission_pct IS NOT NULL);
No 5 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select first_name, last_name, e_id, commission_pct, department_id from employees WHERE department_id='D02' and commission_pct > ALL (Select commission_pct FROM employees WHERE department_id = 'D01' AND commission_pct IS NOT NULL); +-----+-----+-----+-----+-----+ first_name last_name e_id commission_pct department_id +-----+-----+-----+-----+-----+ Jane Smith E002 0.150 D02 Emily Davis E004 0.200 D02 James Taylor E007 0.180 D02 Thomas Lee E011 0.100 D02 +-----+-----+-----+-----+-----+ 4 rows in set (0.002 sec) </pre>

No 6 Query (as Plain Text)	Select department_id, count(e_id) as total_employees FROM employees group by dept_id HAVING MAX(salary) <= 70000;
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select department_id, count(e_id) as total_employees FROM employees group by department_id HAVING MAX(salary) <= 70000; Empty set (0.001 sec) </pre>
No 7 Query (as Plain Text)	select department_id as department_id, job_id,salary from employees e1 WHERE salary < ANY (Select salary FROM employees e2 WHERE e1.department_id = e2.department_id AND e1.job_id != e2.job_id);
No 7 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> select department_id as department_id, job_id,salary from employees e1 WHERE salary < ANY (Select salary FROM employees e2 WHERE e1.department_id = e2.department_id AND e1.job_id != e2.job_id); +-----+-----+-----+ department_id job_id salary +-----+-----+-----+ D01 IT_PROG 65000 D02 SA_REP 55000 D01 IT_PROG 62000 D02 SA_REP 58000 D01 IT_PROG 60000 D03 HR_REP 50000 D02 SA_REP 60000 D04 FI_ACC 52000 D01 IT_PROG 67000 D03 HR_REP 48000 D04 FI_ACC 51000 +-----+-----+-----+ 11 rows in set (0.032 sec) </pre>
No 8 Query (as Plain Text)	Select manager_id from employees WHERE manager_id IS NOT NULL GROUP BY manager_id Having MIN(salary)>= 55000;

No 8 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select manager_id from employees WHERE manager_id IS NOT NULL GROUP BY manager_id Having MIN(salary)>= 55000; +-----+ manager_id +-----+ E010 E011 +-----+ 2 rows in set (0.001 sec) </pre>
No 9 Query (as Plain Text)	Select first_name, last_name, e_id, salary, department_id, commission_pct FROM employees e1 WHERE commission_pct = (Select MAX(commission_pct) FROM employees e2 WHERE e1.manager_id = e2.manager_id OR (e1.manager_id is NULL AND e2.manager_id IS NULL));
No 9 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select first_name, last_name, e_id, salary, department_id, commission_pct FROM employees e1 WHERE commission_pct = (Select MAX(c ommission_pct) FROM employees e2 WHERE e1.manager_id = e2.manager_id OR (e1.manager_id is NULL AND e2.manager_id IS NULL)); +-----+-----+-----+-----+-----+-----+ first_name last_name e_id salary department_id commission_pct +-----+-----+-----+-----+-----+-----+ Emily Davis E004 58000 D02 0.200 Sarah Brown E006 50000 D03 0.150 Thomas Lee E011 80000 D02 0.100 +-----+-----+-----+-----+-----+-----+ 3 rows in set (0.001 sec) </pre>
No 10 Query (as Plain Text)	Select job_id, AVG(salary) as Avg_salary FROM employees GROUP BY job_id HAVING AVG(salary) < (Select AVG(salary) FROM employees);
No 10 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301485]> Select job_id, AVG(salary) as Avg_salary FROM employees GROUP BY job_id HAVING AVG(salary) < (Select AVG(salary) FROM employees) ; +-----+-----+ job_id Avg_salary +-----+-----+ FI_ACC 51500.0000 HR_REP 49000.0000 SA_REP 57666.6667 +-----+-----+ 3 rows in set (0.001 sec) </pre>